

Soho Editors

Technical & Development Approach

July 2026

This document explains how we will build the new Soho Editors platform

1. What We Are Building?

Soho Editors runs two different businesses on one website today: a training school and a freelance talent agency. They have different customers, different workflows, and different needs. Putting them on one website makes both harder to use.

So we are building two websites:

- **The Training website** — people browse courses, pick a date, and book and pay online.
- **The Talent website** — clients browse the freelancer roster and submit work requests; Soho Editors matches them with the right freelancer.

Behind both websites sits one shared system: one place where all the data lives, one set of business rules, and one admin panel for the Soho Editors team. Nothing is duplicated.

Overall System Architecture

The platform is built using a modern, scalable technology stack designed for reliability, performance, and long-term maintainability.

Core Technology Stack

- **Frontend:** React/Next.js (both websites)
- **Backend:** FastAPI (shared backend)
- **Database:** PostgreSQL
- **Infrastructure:** Amazon Web Services (AWS)

AWS Services

- **AWS Amplify or Vercel** – Website (frontend) hosting
- **AWS App Runner / EC2** – Backend hosting
- **Amazon RDS** – Managed PostgreSQL database
- **Amazon S3** – File storage
- **Amazon CloudFront** – Content delivery (CDN)

- **Amazon Route 53** – Domain management
- **Amazon SES** – Email delivery

Third-Party Services

- **PayPal** – Payment processing (retained)
- **Vimeo** – Video hosting (retained)

Design Philosophy

The guiding principle behind every technology choice is to:

- Use proven, well-supported technologies.
- Build custom functionality only where Soho Editors requires unique business logic.
- Retain existing systems that already work well instead of rebuilding them unnecessarily. (PayPal, Vimeo)

Platform Structure

The solution consists of **two independent websites** running on **one shared backend system**.

Training Website

Users can:

- Browse available courses
- Select course dates
- Complete bookings online

Talent Agency Website

Clients can:

- Browse freelancer profiles
- Submit work requests
- Be matched with suitable freelancers by Soho Editors

Shared Platform

Both websites share:

- A single backend containing all business logic
- One PostgreSQL database
- One central administration panel

This approach provides several advantages:

- No duplication of data or business logic
- Each website is tailored specifically to its audience
- Changes made to one website cannot unintentionally affect the other

- Customer accounts remain unified
 - Reporting remains centralized
 - Day-to-day management is handled from one administration system
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Frontend Technologies

Both websites are built using **React/Next.js**, a modern framework widely used for commercial web applications.

It was selected for four primary reasons:

Performance

- Pages load quickly because they are pre-rendered rather than generated from scratch on every request.

Search Engine Optimisation (SEO)

- Search engines can fully read the website.
- Preserves Soho Editors' existing Google rankings during migration.

Responsive Design

- A single codebase supports:
 - Desktop
 - Tablet
 - Mobile
- No separate mobile website is required.

Reusable Design System

Common interface components are built once and reused across both websites, including:

- Buttons
- Forms
- Cards
- Navigation
- Shared UI elements

This ensures:

- Consistent branding
 - Faster development
 - Easier maintenance
 - Reduced duplication
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Backend Technologies

The backend is built using **FastAPI**, a modern Python framework.

It serves as the central engine of the platform by:

- Enforcing all business rules
- Managing permissions and authentication
- Acting as the only layer that communicates directly with the database

Core Responsibilities

The backend manages:

- User authentication and permissions
- Training catalogue
- Course bookings
- Checkout process
- Pricing calculations
- Freelancer management
- Work request workflow
- Contracts
- Projects
- Integrations with external services
 - PayPal
 - Email services

Why FastAPI?

FastAPI was selected because it:

- Delivers excellent performance under heavy load.
- Supports a clean, modular architecture that simplifies future development.
- Automatically generates up-to-date API documentation.
- Allows preservation of existing Data during migration.

Database Technology and Structure

A single **PostgreSQL** database stores information for both websites.

PostgreSQL is a mature, enterprise-grade database trusted for business-critical systems worldwide. It can easily handle complex relationships like

- Courses with multiple scheduled dates
- Multiple bookings per course session
- Guest bookings without user accounts

- Freelancers working on multiple simultaneous projects

A custom PostgreSQL structure models these relationships much more effectively.

Training Website Data

The database stores:

- Categories
- Courses
- Course sessions
- Trainers
- Testimonials
- Resources
- Tutorials

Talent Agency Data

The database stores:

- Freelancer profiles
- Clients
- Work requests
- Job offers
- Contracts
- Projects
- Change requests

Shared Data

Both websites share:

- User accounts
- User roles
- Orders
- Site settings

Data Storage and Security

Database Storage

Business data is stored within the PostgreSQL database hosted on **Amazon RDS**, including:

- User accounts
- Courses

- Bookings
- Freelancer profiles
- Work requests
- Contracts
- Orders

Amazon RDS also provides:

- Automated backups
- Monitoring
- Maintenance
- High reliability

File Storage

Uploaded files are stored separately from the database using Amazon S3.

Security Measures

Sensitive information is protected through multiple layers of security.

Passwords

- Never stored in plain text.
- Stored using one-way cryptographic hashing, making them unreadable even to administrators.

Payment Information

- Credit and debit card details are never stored or processed by the platform.
- All payment processing is handled securely by PayPal.

Secrets and Credentials

Sensitive credentials such as:

- Database passwords
- API keys
- Access tokens

are stored securely as encrypted secrets rather than being embedded directly within the application code.

How Files and Media Will Be Managed

Images, showreels, PDFs, CVs, and course materials are **not stored in the database**. Instead, they are stored in **Amazon S3**, while the database only stores references to their locations.

This approach provides several benefits:

- Keeps the database fast and lightweight.
- Provides secure and cost-effective file storage.
- Allows files to scale independently from the application.

File Delivery

Files are delivered using **Amazon CloudFront**, AWS's global Content Delivery Network (CDN).

Benefits include:

- Faster image and file loading worldwide.
- Reduced server load.
- Consistent performance regardless of visitor location.

For example, users browsing from London or Sydney will automatically receive files from the nearest CloudFront location.

Upload Process

Files are uploaded using a simple workflow:

- Select a file.
- Upload it.
- Associate it with the appropriate:
 - Course
 - Trainer
 - Freelancer
 - Resource

Video Hosting

Video is handled differently.

Rather than hosting videos ourselves:

- Paid tutorials remain hosted on **Vimeo**.
- The platform displays:
 - Video listings
 - Pricing
 - Trailers
- Users are redirected to Vimeo for playback.

This avoids the significant cost and complexity of building and maintaining a custom video streaming platform.

API Architecture and Communication

Both public websites and the admin panel communicate with the shared backend through a secure encrypted API.

Whenever data is requested:

1. A request is sent to the backend.
2. The backend verifies the user's identity.
3. The backend checks their permissions.
4. The requested data is returned only if authorized.

Benefits

Because all applications use the same API:

- Business rules are written only once.
- Booking behaviour remains identical across all platforms.
- There is no risk of different parts of the system behaving inconsistently.
- Future integrations become significantly easier.

API Documentation

FastAPI automatically generates technical API documentation, allowing:

- Faster onboarding for new developers.
- Easier third-party integrations.
- Up-to-date documentation without manual maintenance.

Authentication and Security Approach

Security Measures

The platform includes multiple security layers:

- All communication is encrypted using HTTPS.
- Passwords are stored using one-way hashing.
- Every request is validated by the backend.
- Permissions are enforced on every action.
- The database is inaccessible from the public internet.
- Form submissions are validated to prevent malformed or malicious input.
- Software dependencies are regularly updated and patched.

Even if someone attempts to bypass the website interface, backend permission checks prevent unauthorized access.

User Roles and Permissions

The platform defines six user roles.

Visitors

Can:

- Browse both websites.
- View courses.
- View freelancer profiles.

Training Customers

Can:

- Book courses.
- Book as guests or registered users.
- View their bookings.
- Request course date changes.

Talent Clients

Can:

- Register an account.
- Browse freelancers.
- Submit work requests.
- Track project status.

Freelancers

Can:

- Manage their profiles.
- Update portfolios.
- Manage availability.
- Receive job offers.
- Accept or decline assignments.
- View assigned projects.

Trainers

Can:

- Manage their profile.
- Update their biography.

Administrators

The Soho Editors team has full access to:

- Content management
- Courses
- Bookings
- Work requests
- Freelancer matching
- Contracts
- Users
- Site settings

All permissions are enforced by the backend on every request, preventing unauthorized access even if someone attempts to access hidden URLs directly.

Admin Panel Architecture

A single administration panel manages both websites.

It replaces the existing WordPress dashboard and serves as the primary operational interface for the Soho Editors team.

Training Management

Administrators can:

- Create and edit courses.
- Manage course dates.
- Manage trainers.
- Manage testimonials.
- Manage resources.
- Manage tutorials.
- View bookings.
- Confirm or release provisional bookings.
- Manage discount codes.

Talent Management

Administrators can:

- Manage freelancer profiles.
- Review work requests.
- Search freelancers by:
 - Skills
 - Availability
- Send job offers.

- Track offer acceptance.
- Manage contracts.
- Manage active projects.

General Administration

The admin panel also manages:

- User accounts.
 - Website settings.
 - Google Analytics tracking codes.
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Client Content Management

The client has **complete control over website content**, but **not page layouts**.

This distinction is intentional and protects the long-term quality of the website.

Content They Can Manage

Without developer assistance, the client can update:

- Courses
- Course dates
- Prices
- Trainers
- Testimonials
- Resources
- Tutorials
- Freelancer profiles
- Bookings
- Work requests
- Job offers
- Contracts
- Projects
- Site settings

Page Layouts

Page layouts remain fixed.

For example:

- The **Our Trainers** page always follows the approved design.
- Adding a new trainer automatically updates the page.
- The client can continually add new content without redesigning the page.

Any structural design changes require a formal change request.

Hosting and Deployment Strategy

The platform is hosted entirely within the client's own AWS account.

The development team does **not** host production systems on its own infrastructure.

Hosting Components

- **AWS Amplify/Vercel**– Website hosting
- **AWS EC2** – Backend hosting
- **Amazon RDS** – PostgreSQL database
- **Amazon S3** – File storage
- **Amazon CloudFront** – CDN
- **Amazon Route 53** – DNS management
- **Amazon SES** – Email delivery

Deployment Process

Deployment follows a staged approach:

1. Develop the new platform.
2. Test everything in a staging environment.
3. Leave the existing WordPress website fully operational.
4. Perform final approval.
5. Switch the domain to the new platform.
6. Keep the old website available temporarily for rollback if required.

With this approach, expected downtime is either:

- Zero, or
 - Only a few minutes during DNS switching.
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Scalability Considerations

The platform is designed to scale without major architectural changes.

As demand grows:

- Website hosting can scale independently.
- Database resources can be increased.
- Media delivery automatically scales through CloudFront.
- File storage expands automatically within S3.

Because both websites share one backend, new functionality can be added without duplicating infrastructure.

Unlike WordPress, the platform also avoids plugin-related performance limitations.

Performance Optimisation

Performance is addressed throughout the platform.

Key optimisations include:

- Pre-rendered React/Next.js pages for faster loading.
- Global file caching through CloudFront.
- Database optimisation for high-traffic queries such as:
 - Course listings
 - Freelancer searches
- Elimination of WordPress plugin overhead.

These improvements significantly improve perceived speed and responsiveness.

Backup and Recovery Strategy

Data protection is built into the platform.

Database

Amazon RDS provides:

- Automatic backups.
- Point-in-time recovery.
- Monitoring.

Files

Amazon S3 provides:

- Redundant storage.
- File versioning (where enabled).

Migration Protection

Only after the migrated data has been verified is the new platform launched.

Following launch, the existing website remains available for a defined period, allowing rapid rollback if necessary.

Third-Party Integrations

The platform integrates with four primary third-party services.

PayPal

Handles:

- Course payments.
- Secure payment processing.
- Pay in 3 functionality.

Payment card information never enters the platform.

Vimeo

Continues hosting:

- Paid tutorial videos.

The platform links to Vimeo rather than hosting video directly.

Amazon SES

Handles:

- Booking confirmations.
- Joining instructions.
- Password reset emails.
- System notifications.

Google Analytics & Search Console

Used for:

- Website analytics.
- SEO monitoring.

Tracking codes are managed directly through the admin panel.

Future Scope

Payment workflows for the talent agency (client invoicing and freelancer payouts) remain outside the current scope

Development Workflow and Best Practices

Development follows a structured sequence to minimise risk.

Project Phases

1. Discovery and business analysis.
2. Infrastructure setup.
3. Database design.
4. Project foundations.
5. Admin panel.
6. Content catalogue.
7. User accounts.
8. Training bookings and payments.
9. Talent agency workflow.
10. Data migration.
11. Testing.
12. Production launch.

Best Practices

Throughout development:

- All code is version controlled.
- Changes undergo peer review.
- Rollbacks remain possible.
- Testing mirrors the production environment.
- PayPal integration is configured in sandbox mode from the beginning.

Additional Technologies and Services

The current technology stack is considered sufficient for the project's defined scope.

However, should project requirements evolve during development, additional technologies and services can be evaluated and incorporated where they provide clear business value.